

RESEARCH ARTICLE

Investigating the *in vivo* effect of *Tribulus terrestris* extract in middle cerebral artery occlusion rats using LC-MS-based metabolomics combined with molecular docking

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Abstract

Tribulus terrestris L. fruit (TT) is a traditional Chinese herbal medicine used to treat ischemic stroke (IS). This study aimed to investigate the protective effect of TT extract, named TT15, on middle cerebral artery occlusion (MCAO) rats using metabolomics and molecular docking and find the targets of action and the material basis of TT15 against IS. The results of the infarct volume and neurological defect scores confirmed the efficacy of TT15. Serum metabolomics analysis using LC-MS revealed that model group animals experienced a variety of metabolic disturbances when compared to the sham group. TT15 can restore the MCAO-induced serum metabolite changes by modulating multiple metabolic pathways. Six enzymes were highlighted by the metabolite-reaction-enzyme-gene (M-R-E-G) network analysis, which might be the possible targets for the TT15 against IS. Molecular docking analysis was applied to show the binding affinities between active compounds and these enzymes. The representative docking mode with the lowest binding energy between three compounds and phospholipase A 2 (PLA2) and peroxidase (POD) was displayed by the ribbon binding map. This study profiles the metabolic changes in MCAO-induced IS and investigates the efficacy and the corresponding mechanism of TT15 in the treatment of IS.

KEYWORDS

metabolomics, middle cerebral artery occlusion, molecular docking, *Tribulus terrestris* extract, UHPLC-Q-Orbitrap MS

1 | INTRODUCTION

Ischemic stroke (IS) is the leading cause of brain function lesions, neurological disability, and high mortality rates in patients (Kim et al., 2018), which has caused a great economic and social burden globally. A variety of pathophysiology factors, such as oxidative stress, calcium overload, glutamate-mediated excitotoxicity, and inflammation, can result in IS (Song et al., 2019). Regrettably, so far, except for

aspirin and thrombolytic therapy, no conventional effective therapy for IS has been widely accepted (Zheng et al., 2017). Even though Western medicine has positive therapeutic effects for IS, there are some serious side effects, such as cerebral hemorrhage.

Traditional Chinese medicine (TCM) has the advantages of systematic regulation with multicomponents and multitargets and has a few side effects in disease treatment. *Tribulus terrestris* L. (TT) is an ancient medicinal herb, growing in temperate and tropical climates, which is rich in the chemical composition of biologically active substances and chemical elements, including saponins, flavonoids,

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alkaloids, and other trace elements (Tkachenko et al., 2020). Recent studies have indicated that saponins exhibit a wide range of pharmacological activities, including antiaging, cardiovascular and cerebrovascular system protection, lipid regulation, hypoglycemia, sexual strength, antibacterial, anticancer, and anti-inflammatory (Li et al., 2011). Flavonoids can dilate blood arteries, minimize thrombosis, and regulate blood coagulation and blood lipids to enhance the blood flow condition of patients (Hou et al., 2014; Qu & Yang, 2007). Alkaloids and amino acids play a key role in antifungal activity (Xu et al., 2019). Our previous studies found that gross saponins of TT fruit can fight IS probably by alleviating the neurotoxic effects of excitatory amino acids, regulating the activity of nitric oxide (NO), increasing cerebral blood flow, and improving energy metabolism (Guo et al., 2020; Jiang et al., 2011; Wang et al., 2019). In addition to the gross saponins of TT, we obtained another active site from TT, named TT15, using different separation methods, which showed significant protective effects on middle cerebral artery occlusion (MCAO) brain tissue (Xu et al., 2022). However, the effect of TT15 on serum metabolism is not clear, so we performed this study to investigate the protective effect and corresponding regulation mechanism of TT15 on the serum metabolism of MCAO rats using the metabolomics approach combined with molecular docking analysis.

As a branch of systems biology, metabolomics seeks to characterize the global metabolites in a biological system as well as the related dynamic reactions to changes in internal and external stimuli. Using metabolomics, people can identify specific biomarkers under certain physiological and pathological situations. Recently, there has been increased interest in the metabolomics analysis of biological sample profiling, with the key objective of discovering relevant biological markers implicated in pathological states and therapeutic therapies (Han et al., 2022; Marchev et al., 2021; Wang et al., 2015; Wang et al., 2016).

In this study, we established an MCAO-induced IS rat model and administrated TT15 to rats via tail vein injection. The protective effect of TT15 against IS was evaluated by detecting the neurological score and cerebral infarction volume. LC-MS-based metabolomics was used to monitor serum metabolic changes, screen potential targets, and further demonstrate the *in vivo* effect and related mechanism of TT15 against IS. The metabolite-reaction-enzyme-gene (M-R-E-G) network was established to screen out enzymes closely related to the anti-IS effect of TT15. The interactions between active compounds and the core enzymes were assessed via molecular docking. This study provides a strategy for understanding the material basis and underlying mechanisms of the protective effect of TT15 on IS.

2 | MATERIALS AND METHODS

2.1 | Reagents and materials

HPLC-grade methanol and formic acid were provided by Fisher Scientific (Fair Lawn, NJ, USA). Ultrapure water (resistivity = 18.2 MΩ cm) was prepared using a Milli-Q purification system (Millipore, Billerica,

MA, USA). TT15 was prepared according to our previously reported method (Xu et al., 2022), which is described in the Supporting Information.

2.2 | Animal treatment and drug efficacy evaluation

Adult male Sprague-Dawley rats ($n = 30$, body weight: 250 ± 20 g) were obtained from the Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). The MCAO rat model was established, and TT15 was administered to rats via tail vein injection. The infarct area of the brain tissue and the neurological defect scores were detected to assess drug efficacy. Brain tissues were collected for TTC (2,3,5-triphenyltetrazolium chloride) staining. Serum samples of all animals were collected and stored at -80°C until LC-MS analysis. The detailed procedure is provided in the Supporting Information.

2.3 | Serum sample preparation

An aliquot of 100 μl of serum was deproteinized by adding 300 μl of methanol and vortexed for 30 s. After the mixture was centrifuged at 15,000g for 15 min at 4°C , the supernatant was dried under a nitrogen stream. Then the residue was dissolved in 100 μl of ultrapure water and centrifuged at 15,000g for 15 min at 4°C . Then, the supernatant was transferred to a sample vial for LC-MS analysis. An equal volume of the supernatant from all samples was mixed as a quality control (QC) sample that was used for method validation.

2.4 | UHPLC-MS conditions

Chromatographic separation of the serum metabolites was conducted on a Thermo UHPLC system equipped with a Hypersil GOLD column (2.1×50 mm, $1.9 \mu\text{m}$) at 50°C . The mobile phases were 0.1% formic acid in water (solvent A) and 0.1% formic acid in methanol (solvent B). The flow rate was 0.4 ml/min, and the solvent gradient program was set as follows: 5% B, 0–1 min; 5–100% B, 1–16 min; 100% B, 16–20 min; 100–5% B, 20–22 min; and 5% B, 22–27 min. The injection volume was 2 μl . All samples were stored at 4°C during analysis.

MS detection was conducted using a Q-Orbitrap MS equipped with an electrospray ionization source (Thermo Fisher Scientific, San Jose, CA, USA) operating in both positive- and negative-ion modes. The optimal ionization source conditions were set as follows: capillary voltage was 3.5 kV in positive-ion mode or -3.2 kV in negative-ion mode, sheath gas flow was 50 arb, auxiliary gas flow was 15 arb, sweep gas flow was 2 arb, and capillary temperature was 300°C . The full MS with a resolution of 35,000 was applied to obtain MS data for data preparation. The ddMS2 (Top 5) mode with a resolution of 17,500 was conducted for MS/MS spectrum acquisition. The mass range was m/z 100–1500.

2.5 | Data analysis and biomarker selection

The raw data obtained from LC-MS analysis were processed using the Compound Discoverer 3.1 software package with the following parameters: maximum precursor mass was 1500 Da, and minimum peak count was 1 in the select spectra module. The alignment model is the adaptive curve, the maximum shift was 2 min, and mass tolerance was 10 ppm. Intensity tolerance was 30%, and minimum peak intensity was 1,000,000. Other parameters were set as default. Finally, a data matrix consisting of retention time, accurate m/z values, normalized intensities of each signal, and the sample code was generated.

Multivariate statistical analysis was conducted by importing the data set established earlier into the SIMCA software (version 13.0, Umetrics, Umeå, Sweden). To obtain an overview of all samples and screen out the varied metabolites responsible for the classification among the groups, we constructed a partial least squares-discriminant analysis (PLS-DA) model. The quality of the PLS-DA model was evaluated using parameters R^2 and Q^2 , and a permutation test was used to assess the fittingness of the model. Variable importance prediction (VIP) values were used to filter features at first. The t -test and fold change (FC) values were then used to examine the significant differences in these signals among groups. The compound that met the following criteria, $VIP > 1$, $FC > 1.50$ or < 0.66 , and $P < 0.05$, was considered to be differential metabolites and was selected as biomarkers. Then these biomarkers were annotated by comparing the accurate m/z and the spectra of MS/MS fragments with the records in Human Metabolome Database (HMDB) (Wishart et al., 2022).

2.6 | Metabolic network analysis

Metscape is a Cytoscape plug-in and could visualize and interpret metabolomics data to map the identified metabolites to corresponding proteins or genes based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) records (Gao et al., 2010). The KEGG IDs of biomarkers were imported into the Metscape plug-in to generate the M-R-E-G network. Enzymes with a connectivity degree higher than five were selected as the core targets, which may be regulated by TT15 to influence the changes in metabolites in the treatment of IS.

2.7 | Molecular docking

Molecular docking analysis was performed using AutoDock Tools (version 1.5.6) to understand the binding modes between targets and active compounds in TT15 (Trott & Olson, 2010). The crystal structures of target proteins, phospholipase A2 (PLA2) (PDB ID: 7EZZ), aldehyde dehydrogenase (NAD^+) (ALDH NAD) (PDB ID: 3SZB), lyso-phospholipase (LPL) (PDB ID: 1LCL), peroxidase (POD) (PDB ID: 1QGJ), long-chain fatty acid-CoA ligase (LFACL) (PDB ID: 1ULT), and

sphingomyelin phosphodiesterase (SMPD) (PDB ID: 2DDR) were downloaded from the research collaboratory for structural bioinformatics protein data bank (RCSB PDB) database (Burley & Berman, 2021). The structures of the 20 active compounds identified by our previous study (Table S1) were downloaded from the NCBI PubChem database or drawn in ChemDraw. Before docking, the active chemicals (ligands) and target proteins (acceptors) were subjected to energy minimization, deletion of acceptor water molecules, addition of polar hydrogen atoms, charge, and addition of a magnetic field. All parameters were set by default unless otherwise stated.

3 | RESULTS

3.1 | Brain pathological evaluation

After MCAO modeling, except for the sham group, the rats showed slow movement, and the right front paw could not be straightened when the tail was lifted and suspended. Some rats crawled around the circle and even tilted to the right side, and the severe ones could not walk spontaneously and lost consciousness.

The protective effect of TT15 against IS was evaluated by detecting the neurological score and cerebral infarction volume. As shown in Figure S1A, the rats in the sham group had no neurological abnormalities and received a score of zero. The rats in the model group had the greatest neurological deficit score, which was considerably reduced by TT15 administration.

Brain tissue TTC staining results revealed that (Figure S1C) the brain slices in the sham group were stained dark red uniformly, with no infarct region observed. The MCAO operation produced a spectacular infarct area in the coronal brain slices, which is in line with the highest neurological defect score in this group. TT15 treatment could significantly reduce the cerebral infarct volume proportion (Figure S1B) in the ischemic penumbra of rats ($P < 0.05$). The results demonstrated that the administration of TT15 had a protective effect against MCAO in rats.

3.2 | Metabolomics analysis of serum sample

3.2.1 | Method validation

To monitor and evaluate the stability of the system and ensure the reliability of the experimental data, the QC sample was analyzed five times initially for system equilibration and then injected once every five samples. The consistency of the QC samples throughout the experiment was evaluated. A total of 2500 and 2325 features were retrieved from the raw data acquired in positive-ion mode and negative-ion mode, respectively. After data preprocessing, the resulting data set containing information on all samples was subjected to perform principal component analysis (PCA) for visualizing the distribution of QC samples. As shown in Figure 1, the relatively close grouping of the QC samples at the center of all samples was observed

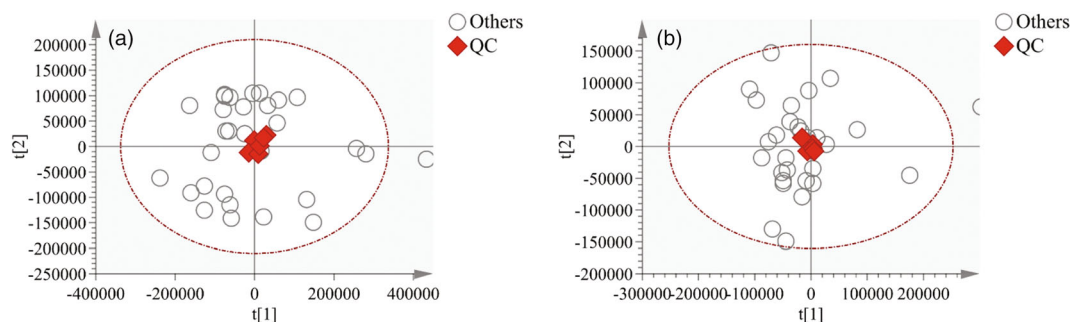


FIGURE 1 The PCA (principal component analysis) score plot of all samples (○) with the highlight of the QC (quality control) (◆) sample in (a) positive-ion mode and (b) negative-ion mode

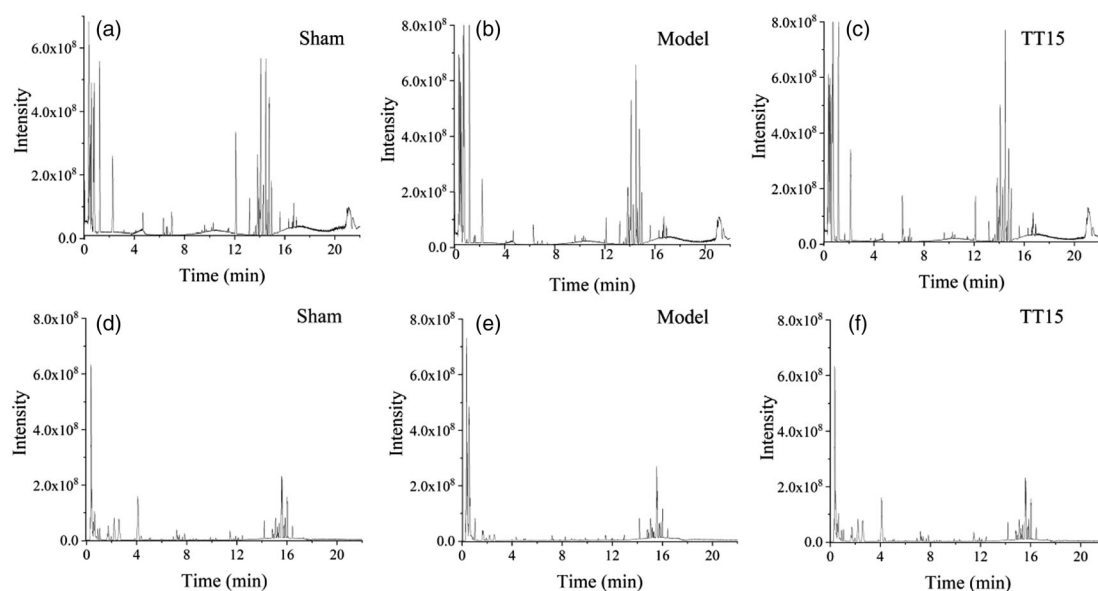


FIGURE 2 The base peak chromatograms (BPCs) of serum from all groups based on the data acquired in (a–c) positive-ion mode and (d–f) negative-ion mode.

in the PCA score plot in both positive- and negative-ion modes, demonstrating the strong repeatability and stability of the proposed LC–MS method.

3.2.2 | Serum metabolite profiling and pattern recognition

Metabolic profiling of serum samples was acquired using a UHPLC–Q–Orbitrap MS instrument operated in both positive- and negative-ion modes. The representative base peak chromatograms of serum samples from the sham group, model group, and TT15 group are shown in Figure 2. The peaks are considered to be chemical fingerprints of serum endogenous metabolites, and their alterations are the reflection of the metabolic perturbation caused by MCAO modeling as well as the protective impact of TT15.

Due to the complexity of the LC–MS spectra, it is a challenge to distinguish differences among groups just via visual inspection of LC–

MS spectra. Therefore, multivariate statistical analysis was used to dig into the minor changes that exist in the metabolite and discover compound changes.

3.2.3 | Multivariate statistical analysis

The PLS–DA model was established to perform pattern recognition as well as to discover the potential biomarkers associated with IS. The parameters of the established PLS–DA models are presented in Table S2, suggesting the established model had great predictability and fitness. As shown in Figure S2, the results of the permutation test ($n = 200$) showed that the value of intercepts R^2 and Q^2 was lower than the original value, indicating no overfitting occurred. As shown in the score plot of PLS–DA (Figure 3a,b), a clear classification between the two groups was observed and located on both sides of the y-axis, indicating that MCAO surgery affected the serum metabolome. After the administration of TT15, the sample point was separated from the

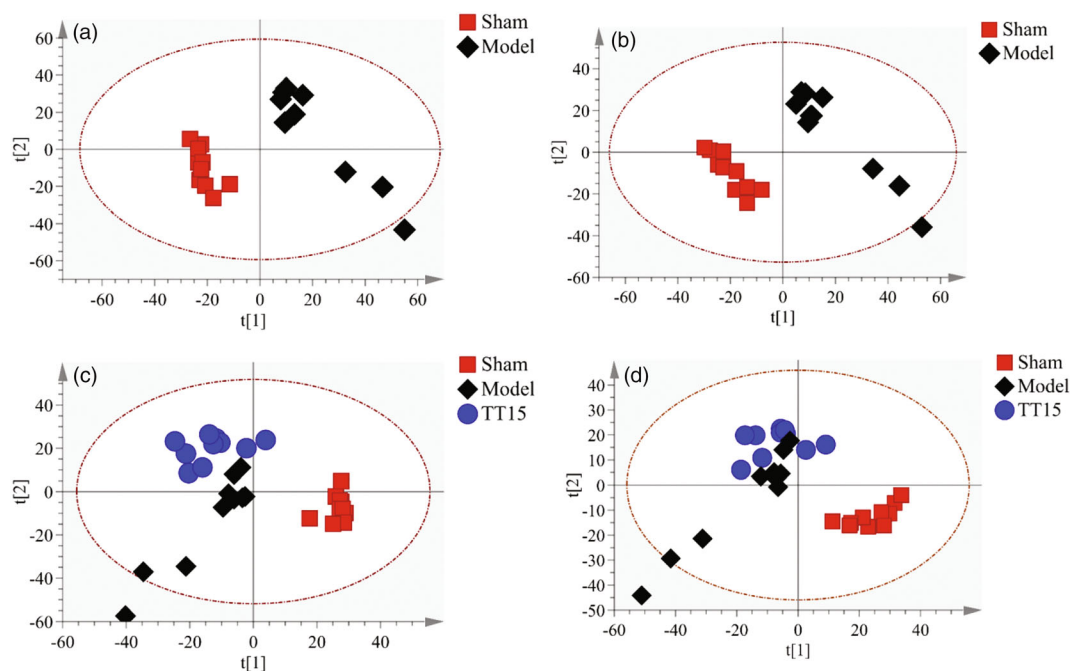


FIGURE 3 PLS-DA (partial least squares-discriminant analysis) score plots of sham group (■), model group (◆), and TT15 group (●) based on the data acquired in (a, c) positive-ion mode and (b, d) negative-ion mode.

model group (Figure 3c,d), which suggested that TT15 regulated the metabolic profile of IS rats and improved their metabolic disorders to skew them toward normal rats.

3.3 | Biomarker annotation and pathway analysis

The differential metabolites with $VIP > 1.0$, $FC > 1.50$ or < 0.66 , and $P < 0.05$ were considered as the biomarkers. Biomarker annotation was achieved by comparing the acquired accurate MS and tandem MS information with database records. Here we take the metabolite LPC (20:1) at retention time (RT) 14.90_m/z 550.3853 as an example to exhibit the metabolite annotation procedures. As shown in Figure 4(a), by comparing the difference between m/z 550.3853 and m/z 572.3684, we assigned these two ions to the protonated ion and the ion of $[M+Na]^+$, respectively. The protonated ion was selected to conduct MS/MS acquisition under the collision energy of HCD40. The tandem MS spectrum is shown in Figure 4(b). The fragment ions at m/z 532.3743 and m/z 184.0726 corresponded to the $[M+H-H_2O]^+$ and $[PC \text{ headgroup}+H]^+$, respectively. The fragment ions at m/z 104.1070 and m/z 124.9994 were assigned to the ions of $[(CH_3)_3N(CH_2)_2O+H]^+$ and $[cyclophosphane \text{ ring}+H]^+$, respectively. The ion at m/z 367.3185 resulted from the loss of the head group from the parent ion. The most likely fragmentation pathway is shown in Figure 4(c). Based on the information provided, this ion was tentatively assigned to LPC (20:1). Finally, 29 differential metabolites related to IS were annotated as biomarkers and are presented in Table 1. The effects of TT15 on these metabolites were next studied by comparing their intensity between the TT15 group and the model

group, which revealed that 11 of them were altered substantially ($P < 0.05$).

Changes in these biomarkers suggest changed metabolic phenotypes, which may provide clues to the mechanisms of TT15 effectiveness. A heat map was created based on the extracted peak intensity to visualize the pattern of these biomarkers in the samples from different groups (Figure 5a). Metabolites with comparable changes in abundance were labeled with similar colors and grouped. The color difference between the sham and model groups suggested metabolic abnormalities after MCAO modeling. The levels of these metabolites were partially restored and close to the color in the sham group after TT15 treatment, suggesting that TT15 recovered MCAO-induced metabolic disturbances. To understand pathways and the potential mechanisms that were related to the TT15 protective effect on IS, the selected biomarkers were subjected to perform metabolic pathway analysis using MetaboAnalyst 5.0 (Pang et al., 2021) (Figure 5b; Table S3). The results showed that biosynthesis of unsaturated fatty acids; phenylalanine, tyrosine, and tryptophan biosynthesis; arginine and proline metabolism; and aminoacyl-tRNA biosynthesis may be greatly attributed to the efficacy of TT15 against IS.

3.4 | Metabolic network analysis and molecular docking

An M-R-E-G network was constructed by importing the biomarkers into Cytoscape to study the detailed regulation mechanism of TT15. As shown in Figure 6, the established network contains 51 metabolites, 30 enzymes, and 81 genes. The topology analysis data

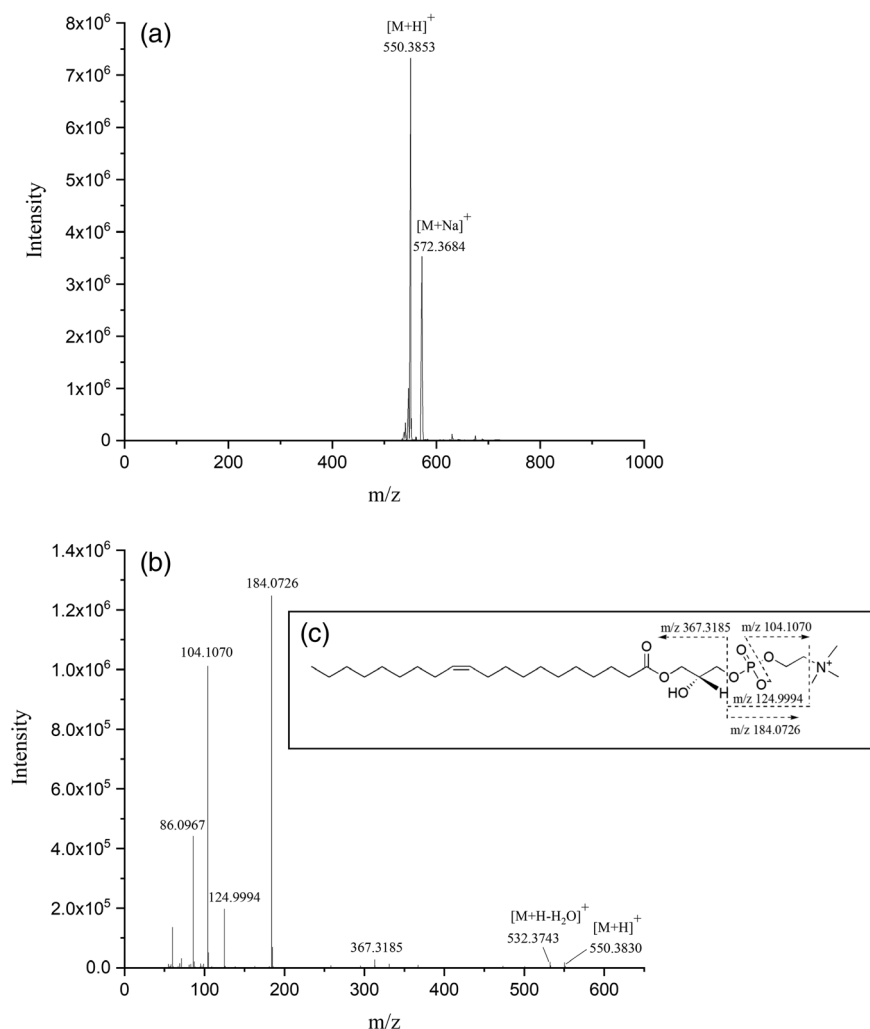


FIGURE 4 (a) MS spectrum, (b) tandem MS spectrum, and (c) the probable fragmentation pathway of the LPC (20:1) at RT 14.90_m/z 550.3853.

(Figure S3) showed that enzymes, including PLA2, ALDH (NAD⁺), LPL, POD, LFACL, and SMPD, had a connectivity degree higher than five, which were greatly involved in the metabolic pathway and were selected as the core targets of TT15 in the treatment of IS.

Molecular docking analysis was performed to obtain the binding affinity between active compounds and core targets. Twenty sets of compound–protein docking were generated and are presented in Table S1. Lower binding energy was observed between compounds and PLA2 or POD, for example, C9-PLA2 (−7.75 kcal/mol), C4-PLA2 (−7.62 kcal/mol), C1-PLA2 (−7.16 kcal/mol), C9-POD (−8.29 kcal/mol), C4-POD (−7.48 kcal/mol), and C1-POD (−6.81 kcal/mol), indicating that the interaction between these three compounds and enzymes may play an important role in TT15 when treating MCAO rats. The binding modes of C9, C4, and C1 with PLA2 and POD are shown in Figure 7.

4 | DISCUSSION

In this study, the changes in endogenous serum metabolites were analyzed using LC–MS-based metabolomics to investigate the protective effect of TT15 on MCAO-induced IS and related mechanisms. Our

research revealed that TT15 provides protective effects against IS by reducing the infarct volume and alleviating neurological deficits. The PLS-DA score plot showed that the metabolic profile of the serum of TT15 rats showed separation trends from the model group although small overlaps were observed, suggesting TT15 corrected IS metabolic disorders by reversing the unbalanced metabolites and promoting metabolic regression to the normal range. The pathway analysis showed that these protective effects were greatly involved in energy metabolism, amino acid metabolism, fatty acid metabolism, and purine metabolism. The M-R-E-G network analysis revealed that PLA2 and POD were significantly connected with the biomarkers screened out in the metabolomics investigation, indicating that two enzymes are likely candidate targets for the protective effect of TT15 against MCAO. The results of molecular docking showed that C9, C4, and C1 have a high affinity with PLA2 and POD, which suggests that these three active compounds played a key role in the treatment of IS by regulating these enzymes.

It was reported that valine significantly decreased the expression of interleukin-6, and cyclooxygenase-2 in lipopolysaccharide (LPS)-activated RAW 264.7 macrophages, resulting in an anti-inflammatory effect. Valine reduced inducible nitric oxide synthase (iNOS) expression, which prevented LPS-induced NO generation (Lee et al., 2017).

TABLE 1 Potential biomarkers and their changes in different groups.

Number	Metabolite	Chemical formula	Retention time	Detected m/z	Adducts	Library ID	Trends ^a	Trends ^b
1	Palmitoleic acid	C ₁₆ H ₃₀ O ₂	15.37	253.2172	[M-H] ⁻	HMDB0003229	↑###	↓
2	Uric acid	C ₅ H ₄ N ₄ O ₃	0.43	335.0498	[2M-H] ⁻	HMDB0000289	↑#	↓*
3	Diethyl fumarate	C ₈ H ₁₂ O ₄	5.50	171.0656	[M-H] ⁻	HMDB0038601	↑#	↓
4	Stearic acid	C ₁₈ H ₃₆ O ₂	16.45	283.2643	[M-H] ⁻	HMDB0000827	↑###	↓
5	Ascorbic acid 2-sulfate	C ₆ H ₈ O ₉ S	0.40	255.9889	[M-H] ⁻	HMDB0060649	↑#	↓
6	L-Phenylalanine	C ₉ H ₁₁ NO ₂	1.06	254.9817	[M-H] ⁻	HMDB0000159	↑###	↓*
7	3-Hydroxymethylglutaric acid	C ₆ H ₁₀ O ₅	0.64	161.0449	[M-H] ⁻	HMDB0000355	↑##	↓*
8	Guanidoacetic acid	C ₃ H ₇ N ₃ O ₂	0.63	116.0988	[M-H] ⁻	HMDB0000128	↑#	↓
9	Dihomo-α-linolenic acid	C ₂₀ H ₃₄ O ₂	15.87	305.2490	[M-H] ⁻	HMDB0060039	↑#	↓
10	Docosapentaenoic acid (22n-3)	C ₂₂ H ₃₄ O ₂	15.77	329.2488	[M-H] ⁻	HMDB0006528	↑##	↓
11	Myristic acid	C ₁₄ H ₂₈ O ₂	15.16	228.2087	[M-H] ⁻	HMDB0000806	↑###	↓
12	16-Hydroxyhexadecanoic acid	C ₁₆ H ₃₂ O ₃	13.89	227.2014	[M-H] ⁻	HMDB0006294	↑##	↓
13	Goshuyic acid	C ₁₄ H ₂₄ O ₂	14.11	223.1701	[M-H] ⁻	HMDB0000560	↑###	↓*
14	Dimethyl fumarate	C ₆ H ₈ O ₄	0.62	143.0342	[M-H] ⁻	HMDB0031257	↑###	↓**
15	Dimethylguanidino valeric acid	C ₈ H ₁₅ N ₃ O ₃	8.32	199.0971	[M-H] ⁻	HMDB0240212	↑##	↓**
16	Myristoleic acid	C ₁₄ H ₂₆ O ₂	14.67	225.1858	[M-H] ⁻	HMDB0002000	↑###	↓
17	Cholic acid	C ₂₄ H ₄₀ O ₅	12.04	407.2810	[M-H] ⁻	HMDB0000619	↑#	↑
18	Cyclo	C ₂₇ H ₄₆ N ₆ O ₇	14.67	564.3306	[M-H] ⁻	HMDB0303709	↓##	↑
19	Dimethyl fumarate	C ₆ H ₈ O ₄	0.62	145.0495	[M+H] ⁺	HMDB0031257	↑###	↓*
20	Homosalate	C ₁₆ H ₂₂ O ₃	11.92	263.1616	[M+H] ⁺	HMDB0253209	↑###	↓*
21	PC (18:1-O(12,13)/16:0)	C ₄₂ H ₈₀ NO ₉ P	16.43	775.5739	[M+H] ⁺	HMDB0285954	↑###	↓*
22	L-Valine	C ₅ H ₁₁ NO ₂	0.35	118.0864	[M+H] ⁺	HMDB0000883	↓##	↑*
23	Boldione	C ₁₉ H ₂₄ O ₂	11.44	285.1845	[M+H] ⁺	HMDB0003422	↑#	↓
24	Leucylproline	C ₁₁ H ₂₀ N ₂ O ₃	0.74	229.1544	[M+H] ⁺	HMDB0011175	↑#	↓
25	γ-Aminobutyric acid	C ₄ H ₉ NO ₂	0.40	229.1181	[2M+Na] ⁺	HMDB0000112	↑#	↓
26	LPC (20:3)	C ₂₈ H ₅₂ NO ₇ P	14.14	546.3548	[M+H] ⁺	HMDB0010393	↓##	↑
27	SM (d18:1/20:0)	C ₄₃ H ₈₇ N ₂ O ₆ P	17.09	781.6187	[M+Na] ⁺	HMDB0012102	↓##	↑
28	α-Linolenic acid	C ₁₈ H ₃₀ O ₂	14.23	279.2319	[M+H] ⁺	HMDB0001388	↑##	↓
29	LPC (20:1)	C ₂₈ H ₅₆ NO ₇ P	14.90	550.3853	[M+H] ⁺	HMDB0010391	↑##	↓**

Note: Compared with the peak areas, the compound showed upregulated trends (↑) and downregulated trends (↓).

^aChange trend of model group versus sham group (#*P* < 0.05, ##*P* < 0.01, and ###*P* < 0.001).

^bChange trend of TT15 group versus model group (**P* < 0.05, ***P* < 0.01).

Moreover, the valine deficiency resulted in the disturbed function of the central nervous system in rats, and the limbs trembled in the form of ataxia (Guo et al., 2019). Valine is an essential amino acid that belongs to the branched-chain amino acids (BCAAs). BCAAs act as an energy source and substrate to enlarge the intermediate pool of the tricarboxylic acid (TCA) cycle as well. It has been reported that IS can cause energy metabolism disorder (Liu et al., 2014). When an IS occurs, the energy supply to the brain is insufficient due to the slow-down of the TCA cycle. The BCAAs can affect the intermediates of the TCA cycle via forming glutamate or branched-chain acyl-coenzyme A, thus affecting energy metabolism (Biswas et al., 2019). Valine was closely associated with inflammatory response, improved energy metabolism, and neuroprotective functions. Compared with the sham group, L-valine decreased in the model group, suggesting

energy disorder, inflammatory reaction, and nervous system dysfunction appeared during MCAO modeling. TT15 exerted anti-IS effects mainly by regulating valine metabolism.

Uric acid is the ultimate product of purine metabolism, which is mainly generated in the liver and excreted through the kidney (Jang et al., 2020). Several epidemiological studies have suggested a positive relationship between elevated serum uric acid levels and an increased risk of cardiovascular events, such as coronary heart disease, stroke, and heart failure (Hozawa et al., 2006; Keenan et al., 2016; Kim et al., 2009). Even after controlling for other risk factors, elevated uric acid remained an independent risk factor for cardiovascular events in type 1 diabetes patients with hypertension, diabetes, and preexisting cerebrovascular and cardiovascular illness (Pilemann-Lyberg et al., 2019). But it is unclear whether elevated uric acid levels are a

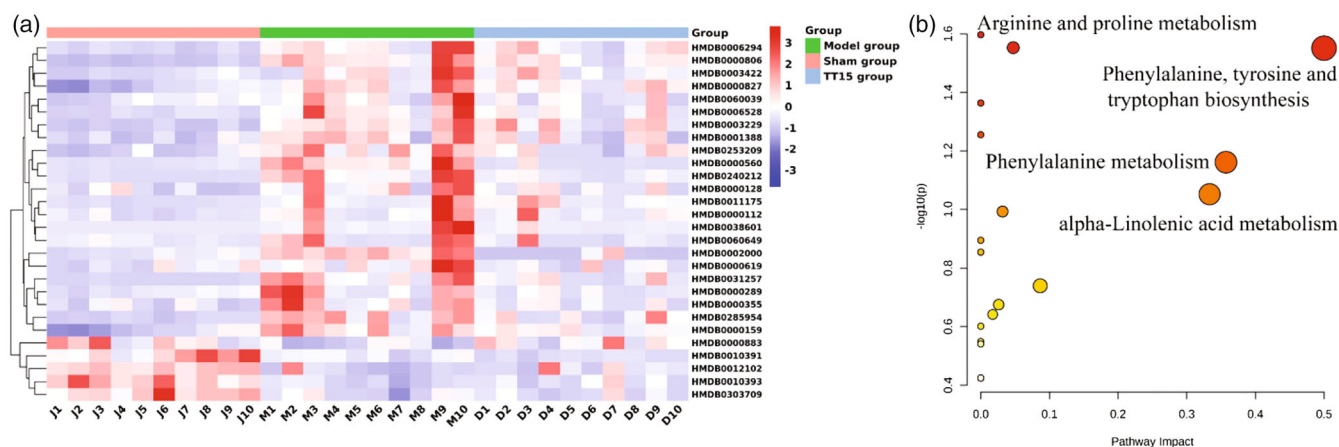


FIGURE 5 (a) The heat map of the differential metabolites from serum. Rows: metabolites; columns: samples. On the top is the cluster of samples, and on the right is the cluster of metabolites. Red means the metabolites are expressed at a higher level, and blue means the metabolites are expressed at a lower level. (b) The result of pathway analysis is visualized by bubble plots.

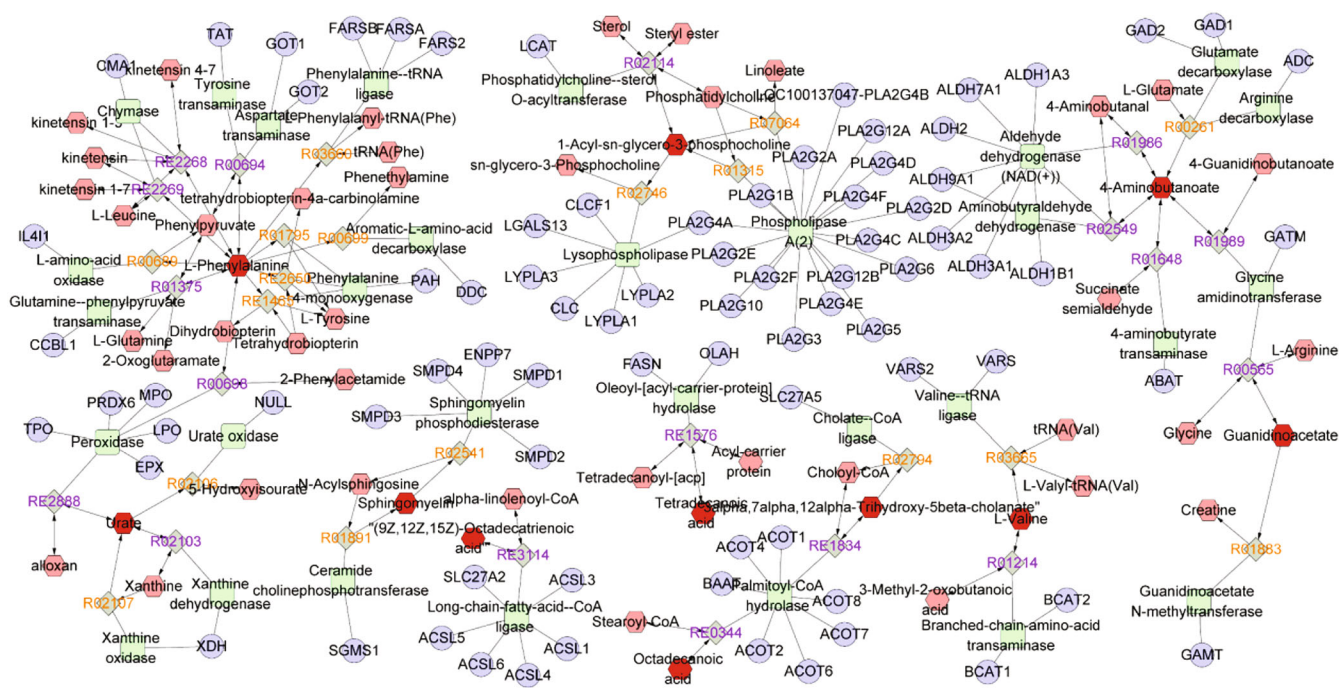


FIGURE 6 The metabolite-reaction-enzyme-gene networks based on the biomarkers screened metabolomics analysis. The faint red hexagons (), gray diamonds (), green rectangles (), scarlet hexagons (), and purple circles () represent the metabolites, reactions, enzymes, biomarkers, and genes, respectively.

cause or a consequence of IS. In comparison to the Sham group, uric acid levels were increased in the model group, whereas that in the TT15 group was downregulated to a similar level in the sham group.

TT15 contains numerous active compounds that synergistically perform therapeutic functions. Six enzymes that are highly connected to the biomarkers were selected using the M-R-E-G network analysis, suggesting that these enzymes might be a viable target for TT15 to exert anti-MCAO efficacy. Thus, 20 compounds in TT15 and these target proteins were selected to conduct molecular docking analysis. The results showed that PLA2 and POD have good binding energy

with the active compounds. PLA2 is a superfamily of enzymes that may play a major role in inflammatory response. Elevated free radicals during IS initiate the hydrolysis of phospholipids via the activation of PLA2 and then release the lysophospholipids and free fatty acids (FFAs), for example, arachidonic acid, which is a metabolic precursor of pro-inflammatory eicosanoids. After IS, FFAs can accumulate and undergo oxidative metabolism via nonenzymatic and enzymatic processes mostly mediated by cyclooxygenases (COXs) and lipoxygenases (LOXs), leading to the generation of lipid oxoderivatives that regulate inflammatory and pro-oxidative processes (Farina

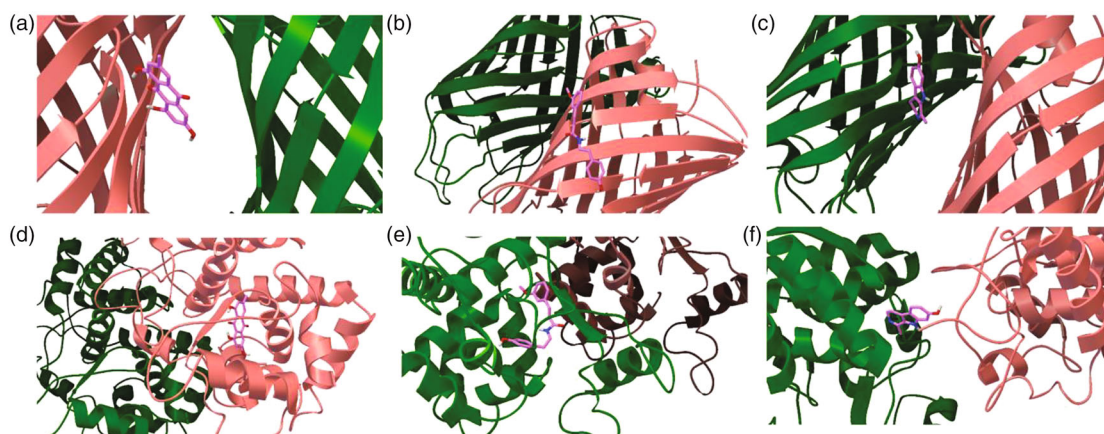


FIGURE 7 The 3D ribbon binding map of (a–c) C9, C4, and C1 to PLA2 (phospholipase A2) and (d–f) C9, C4, and C1 to POD (peroxidase).

et al., 2021). Both COXs and LOXs have been identified as possible pharmacological targets for stroke treatment (Klaus, 2013; Lake et al., 2017; Liu et al., 2017; Yang et al., 2020).

PODs are heme-containing enzymes found in bacteria, fungi, plants, and mammals, which can catalyze the reduction of hydrogen peroxide and the oxidation of various organic and inorganic substrates (Lazzarotto et al., 2021). Myeloperoxidase is a major inflammatory POD enzyme that is secreted by activated neutrophils and macrophages to participate in systemic inflammation response as inflammatory mediators. Inflammatory infiltration is linked to the formation of atherosclerotic plaques and their complications (Vivekanandan-Giri et al., 2013). In patients with soft and complex plaques, inflammatory factors are more active, and plaques are more unstable and prone to fall off to cause IS. Glutathione peroxidase 4 (GPX4) is one of the known PODs in mammals and is the main reactant in the body against oxidation reaction. It prevents the peroxide action of free radicals from damaging cells by reducing lipid peroxides. The increase in oxidative stress reaction and peroxidation products is an important physiological and pathological process of IS (Li et al., 2020). The reduction in GPX4 directly responded to the enhancement of the redox reaction in the body, which eventually contributed to the onset and progression of IS (Bao et al., 2021; Qi et al., 2022).

The results of molecular docking analysis showed interactive modes between the active compounds and the target proteins. C9, C4, and C1 have substantial affinities for PLA2 and POD, with bonding energies less than -6.8 kcal/mol, indicating that these three compounds may protect against MCAO by modulating these two targets. However, their upstream regulatory genes and downstream effector molecules were not fully investigated in the present study. And their synergistic effects must also be studied in depth.

5 | CONCLUSION

In the present study, we used LC–MS–based metabolomics combined with molecular docking to investigate the protective effect of TT15 against IS. The results showed that there were 29 metabolites with

prominent changes in the model group compared with the sham group. These biomarkers were associated with energetic metabolism, amino acid metabolism, fatty acid metabolism, and purine metabolism. After treatment with TT15, 11 of them were reversely regulated significantly. The M-R-E-G network analysis showed six enzymes with a connectivity degree higher than five, which was probably linked to the action of TT15. The binding energy between the active compounds and targets was analyzed using molecular docking. C9, C4, and C1 exhibited a lower docking affinity with PLA2 and POD and may exert mainly protective effects on MCAO. Integrating metabolomics and molecular docking may provide better guidance for investigating the mechanism of action of TCM in disease treatment and screening the material basis of TCM in the treatment of IS. In-depth validation studies are needed in the future to better understand the regulating mechanism of TT15 and to aid in the development of TT15 as a viable additional medicine for the treatment of IS.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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